Module 11: References

11.1 What is a Reference

- A reference is another name (alias) for an existing variable
- Once created, a reference always refers to the same variable and cannot be changed to refer to another
- Must be initialized at the time of declaration

```
int x = 10;
int &ref = x; // ref is an alias for x
ref = 20;     // changes x to 20
```

11.2 References vs Pointers

- A reference cannot be null; a pointer can be null
- A reference is automatically dereferenced; a pointer requires * to access the value
- A reference cannot be reassigned to refer to a different variable
- References are simpler and cleaner for most use cases
- Pointers offer more power: pointer arithmetic, dynamic memory, reassignment

11.3 References in Functions

- Pass by reference: the function receives an alias to the original variable

```
void swap(int &a, int &b) {
int temp = a;
a = b;
b = temp;
}
swap(x, y); // x and y are actually swapped in the caller
```

- const reference: allows reading the value without copying, while preventing modification

```
void print(const string &name) {
cout << name;
// name = "other"; // error: cannot modify a const reference
}
```